

CHAPTER-WISE PROBLEMS

Principal Component Analysis & Standard Deviation Control
MIT/Stanford Level

Statistical Foundation of Data Science
CSU1658

December 2025

Contents

1	Principal Component Analysis (PCA)	2
2	Controlling Standard Deviation and Variance	5

1 Principal Component Analysis (PCA)

Problem 1: Stock Portfolio Dimensionality Reduction

An investment firm tracks 4 stock indices over 5 trading days. The centered return data (%) is:

Day	Tech	Energy	Finance	Healthcare
1	1.2	-0.5	2.1	-1.8
2	-0.8	0.3	-1.5	1.2
3	2.3	-1.1	3.8	-3.2
4	-1.5	0.7	-2.6	2.2
5	0.4	-0.2	0.9	-0.8

PCA results: Eigenvalues are $\lambda_1 = 12.85$, $\lambda_2 = 3.42$, $\lambda_3 = 1.18$, $\lambda_4 = 0.35$. First PC loading vector: (0.48, -0.22, 0.75, -0.63).

- Compute the 4×4 covariance matrix using the formula $\mathbf{C} = \frac{1}{n-1} \mathbf{X}^T \mathbf{X}$ where $n = 5$ days. Verify symmetry and identify which pair of indices shows the strongest correlation from the off-diagonal elements.
- Using the given eigenvalues, verify they are in descending order. Calculate the condition number $\kappa = \lambda_1/\lambda_4$. What does this indicate about the stability of the covariance matrix?
- Calculate the proportion of variance explained by each PC. How many components are needed to retain 90% of total variance? Using the elbow method, where would you truncate?
- Project the original data onto the first two PCs. Calculate the reconstruction error (Frobenius norm squared) when using only 2 components versus the total variance. What percentage of information is lost?
- Interpret the first PC loading vector. Which index contributes most to PC1? If PC1 score increases by 1 unit, how does the Tech Index return change? What economic relationship does the sign pattern (positive for Tech/Finance, negative for Energy/Healthcare) suggest about market sectors?
- Calculate the Mahalanobis distance from origin to the point (1.5, -0.8, 2.5, -2.0) in both the original 4D space and the PC space using all 4 components. Verify that PCA preserves this distance measure.

Problem 2: Image Compression with PCA

A grayscale 6×6 image patch (pixel intensities 0-255) is mean-centered for PCA compression. The image shows a lighting gradient from dark (rows 1-2, intensity 45) to medium (rows 3-4, intensity 120) to bright (rows 5-6, intensity 195).

SVD analysis yields singular values: $\sigma_1 = 523.4$, $\sigma_2 = 8.7$, $\sigma_3 = 3.2$, $\sigma_4 = 1.8$, $\sigma_5 = 0.9$, $\sigma_6 = 0.3$.

Reconstruction quality:

k	Storage (values)	Compression Ratio	Reconstruction Error (%)
1	13	2.77:1	2.8
2	26	1.38:1	1.2
3	39	0.92:1	0.8
6 (full)	78	0.46:1	0.0

- Verify the relationship between singular values and eigenvalues using $\lambda_i = \sigma_i^2/(n-1)$ where $n = 6$. Calculate the Frobenius norm squared $\|\mathbf{X}\|_F^2$ and confirm it equals $\sum_{i=1}^6 \sigma_i^2$.
- For each $k=1,2,3$, calculate the compression ratio comparing original storage (36 values) to compressed storage ($k \times 13$ values for PC scores, loadings, and singular values). Which k achieves reconstruction error below 5% with maximum compression?
- The first PC captures the lighting gradient. To remove this artifact, project the image onto the subspace orthogonal to PC1 using $\mathbf{P}_\perp = \mathbf{I} - \mathbf{u}_1 \mathbf{u}_1^T$. Calculate the Frobenius norm before and after this de-lighting operation. What percentage of energy is removed?
- Storage break-even analysis: PCA becomes storage-efficient when $k(m+n+1) < mn$. For $m = n = 6$, solve for maximum k . Repeat for a 100×100 image. At what k does PCA become inefficient?
- Scale to 1000 patches of 6×6 pixels. If global PCA on all patches finds the first 3 PCs explain 95% variance, calculate: (a) total storage for raw data, (b) storage using 3-component PCA, (c) overall compression ratio.
- White noise with variance $\sigma_n^2 = 25$ contaminates the image. Using Marcenko-Pastur threshold $\sigma_{opt} = \sigma_n \sqrt{m}(1 + \sqrt{n/m})$ with $m = n = 6$, determine which singular values exceed the noise floor and should be retained. Compare σ_{opt} to each σ_i .

Problem 3: Gene Expression Analysis with High-Dimensional PCA

A genomics study measures expression levels of $p = 1000$ genes across $n = 50$ patients (20 healthy, 30 diseased). The standardized data matrix $\mathbf{X}_{50 \times 1000}$ has zero mean and unit variance columns.

PCA eigenvalue results:

Component	Eigenvalue	Variance Explained (%)	Cumulative (%)
PC1	125.3	12.5	12.5
PC2	87.6	8.8	21.3
PC3	62.4	6.2	27.5
PC4	45.8	4.6	32.1
PC5	38.2	3.8	35.9
PC6-50	varies	64.1	100.0

Sum of all 50 non-zero eigenvalues: 1000 (by standardization). PC1 separates healthy vs diseased with 85% classification accuracy.

Cross-validation reconstruction errors:

k components	CV Error	Std. Error	Relative to k=1
1	15200	850	1.00
2	9800	620	0.64
3	7200	490	0.47
5	5400	380	0.36
10	4100	320	0.27

- In the high-dimensional regime ($p = 1000 \gg n = 50$), the covariance matrix $\mathbf{X}^T \mathbf{X}$ is 1000×1000 but rank-deficient. Calculate its rank. How many eigenvalues are exactly zero? Explain why we can equivalently perform PCA using the 50×50 Gram matrix $\mathbf{X}\mathbf{X}^T$.
- Calculate cumulative variance explained by first 5 PCs. Under the Marcenko-Pastur null (random matrix theory), the expected largest eigenvalue is $\lambda_{\max}^{\text{null}} = 49(\sqrt{p/n} + 1)^2 \approx 1465$ for random data. Compare this to the observed $\lambda_1 = 125.3$. Does this suggest meaningful structure or random noise?
- Sparse PCA with L1 regularization identifies 50 genes (5% sparsity) with non-zero PC1 loadings. The top 5 gene loadings are: 0.18, 0.15, 0.14, 0.12, 0.11. Calculate the participation ratio $PR = (\sum |w_i|)^2 / \sum w_i^2$ to measure effective dimensionality. How many genes are "effectively" contributing?
- Permutation test: Disease labels are shuffled 1000 times and PC1 separation (ROC AUC) recalculated each time. The observed AUC of 0.85 is exceeded in only 12 permutations. Calculate the empirical p-value. Is PC1 significantly associated with disease at $\alpha = 0.05$?
- Using the cross-validation results table, apply the one-standard-error rule to select the optimal number of PCs: choose the smallest k whose CV error is within 1 SE of the minimum. Calculate SE thresholds and determine the optimal k. Justify whether k=3 or k=5 is preferable.
- Kernel PCA using polynomial kernel $K(\mathbf{x}_i, \mathbf{x}_j) = (1 + \mathbf{x}_i^T \mathbf{x}_j)^2$ captures non-linear interactions. After centering, the kernel matrix has eigenvalues $\mu_1 = 18500$, $\mu_2 =$

12300, with trace 250000. Calculate the variance explained by the first kernel PC (7.4%). Compare to linear PCA's first PC (12.5%). What does the lower variance in kernel space suggest about gene interaction patterns?

2 Controlling Standard Deviation and Variance

Problem 1: Quality Control in Manufacturing Process

A semiconductor manufacturer monitors wafer thickness (in micrometers) to maintain quality. Target specifications: mean $\mu_0 = 500\mu m$ with standard deviation $\sigma_0 = 2\mu m$. A random sample of $n = 25$ wafers yields:

Sample statistics: $\bar{x} = 500.04\mu m$, $s^2 = 2.87$, $s = 1.69\mu m$

Critical values for testing: $\chi_{0.05,24}^2 = 36.415$, $\chi_{0.025,24}^2 = 39.364$, $\chi_{0.975,24}^2 = 12.401$, $F_{0.05,24,24} = 1.984$

- Test if process variance has increased from target $\sigma_0^2 = 4$. Calculate the chi-square test statistic $(n-1)s^2/\sigma_0^2$ and compare to the critical value. Should the null hypothesis be rejected at $\alpha = 0.05$? What does this mean for manufacturing quality?
- Construct a 95% confidence interval for σ^2 using the chi-square distribution. Calculate both bounds. Does this interval contain the target variance $\sigma_0^2 = 4$? Is the process within acceptable limits?
- The manufacturer wants to detect when σ exceeds $2.5\mu m$ (25% increase) with power 0.80. Using $z_{0.05} = 1.645$ and $z_{0.20} = 0.84$, calculate required sample size from the formula $n \approx \frac{1}{2}[(z_\alpha + z_\beta)/\ln(\sigma_1/\sigma_0)]^2$. Round up to ensure adequate power.
- Moving range control chart: Calculate $MR_i = |x_i - x_{i-1}|$ for consecutive measurements. The average moving range is $\overline{MR} = 1.92$. Using the constant $d_2 = 1.128$ for $n = 2$, estimate process standard deviation $\hat{\sigma} = \overline{MR}/d_2$. Calculate upper control limit: $UCL = 3.267 \times \overline{MR}$. If 3 consecutive MR values exceed UCL, what action should be taken?
- Variance reduction test: A new stabilization technique is proposed. Current process: $n_1 = 25$, $s_1^2 = 2.87$. New process: $n_2 = 25$, $s_2^2 = 1.92$. Perform F-test for $H_0 : \sigma_1^2 = \sigma_2^2$ vs $H_1 : \sigma_1^2 > \sigma_2^2$. Calculate F-statistic and compare to critical value. If significant, what is the percentage reduction in standard deviation?
- Economic analysis: Defective wafers (thickness outside $\mu \pm 3\sigma$) cost \$450 each. With current $\sigma = 1.69$, defect rate is 0.27%. New process reduces σ to 1.39 (same rate but tighter control). For 10,000 wafers per day over 2 years (500 working days), calculate total savings. Is a \$50,000 equipment investment justified?

Problem 2: Portfolio Risk Management and Variance Reduction

A quantitative fund manages a portfolio of $p = 10$ assets. The returns covariance matrix has eigenvalues:

Component	Eigenvalue
λ_1	0.045
λ_2	0.028
λ_3	0.019
λ_{4-10}	0.01 (each)

Current equal-weighted portfolio ($w_i = 0.1$ for all assets) has: - Portfolio variance: $\sigma_p^2 = 0.0089$ (so $\sigma_p = 9.43\%$ annual) - Minimum variance portfolio (MVP) achieves: $\sigma_{MVP}^2 = 0.0062$ - Average pairwise correlation: $\bar{\rho} = 0.35$ - Average individual variance: $\bar{\sigma}^2 = 0.012$

Risk contributions: Asset 1 has $MRC_1 = 0.15$, Asset 2 has $MRC_2 = 0.08$

Inverse covariance vector: $\Sigma^{-1}\mathbf{1} = (12.3, 8.7, 15.2, 6.8, 11.4, 9.6, 13.1, 7.9, 10.5, 8.2)^T$

- The minimum variance portfolio solution is $\mathbf{w}_{MVP} = \Sigma^{-1}\mathbf{1}/(\mathbf{1}^T\Sigma^{-1}\mathbf{1})$. Given $\sigma_{MVP}^2 = 0.0062$, calculate the percentage reduction in volatility from equal-weighted portfolio. What is the new annualized standard deviation?
- Diversification benefit formula: $\sigma_p^2 = \bar{\sigma}^2/n + (1 - 1/n)\bar{\rho}\bar{\sigma}^2$. Verify this equals the given portfolio variance using $n = 10$, $\bar{\rho} = 0.35$, $\bar{\sigma}^2 = 0.012$. As $n \rightarrow \infty$, show that $\sigma_p^2 \rightarrow \bar{\rho}\bar{\sigma}^2$, proving systematic risk cannot be diversified away.
- Risk parity allocates capital so each asset contributes equally: $w_i \times MRC_i = \sigma_p/p$ for all i . The marginal risk contribution is $MRC_i = [\Sigma\mathbf{w}]_i/\sigma_p$. Given $MRC_1 = 0.15$ and $MRC_2 = 0.08$, derive the optimal weight ratio w_1/w_2 . Using the inverse covariance vector, calculate risk parity weights for all 10 assets.
- Value at Risk (VaR): For normal returns, 95% daily VaR is $\mu_p - 1.645\sigma_p$. Given daily expected return 0.08% and daily $\sigma_p = 1.2\%$, calculate 95% and 99% VaR ($z_{0.01} = 2.33$). For a \$10 million portfolio, what is the maximum expected one-day loss at 95% confidence?
- Conditional VaR (CVaR or Expected Shortfall) measures average loss beyond VaR: $CVaR_{0.05} = \mu_p - \sigma_p \times \phi(1.645)/0.05$ where $\phi(1.645) = 0.103$ (standard normal PDF). Calculate CVaR and compare to VaR. Explain why CVaR is more conservative for risk budgeting.
- Dynamic variance targeting maintains constant volatility $\sigma^* = 8\%$ by adjusting leverage: $Leverage_t = \sigma^*/\hat{\sigma}_t$. Volatility is estimated using EWMA: $\hat{\sigma}_t^2 = 0.94\hat{\sigma}_{t-1}^2 + 0.06r_{t-1}^2$. If previous volatility was $\hat{\sigma}_{t-1} = 9.5\%$ and yesterday's return was $r_{t-1} = -2.1\%$, calculate updated volatility and required leverage adjustment. By what factor should positions be scaled?

Problem 3: Experimental Design and Variance Partitioning

A pharmaceutical company tests a new drug across 3 dosage levels with 4 patients per level ($n = 12$ total). Response times (minutes to symptom relief):

Dosage Mean	Patient 1	Patient 2	Patient 3	Patient 4
Low 48.75	45	52	48	50
Medium 39.00	38	42	40	36
High 29.00	28	32	30	26

Grand mean: $\bar{y}_{..} = 38.92$

Critical value: $F_{0.05,2,9} = 4.26$

Two-way ANOVA results (with age blocking): $SS_{\text{Dosage}} = 756$, $SS_{\text{Age}} = 124$, $SS_{\text{Error}} = 82$

Sample size calculation constants: $z_{0.025} = 1.96$, $z_{0.20} = 0.84$

- Perform one-way ANOVA variance decomposition: $SST = SSB + SSW$. Calculate total sum of squares $SST = \sum \sum (y_{ij} - \bar{y}_{..})^2$, between-group $SSB = 4 \sum (\bar{y}_i - \bar{y}_{..})^2$, and within-group $SSW = \sum \sum (y_{ij} - \bar{y}_i)^2$. Verify the partition holds.
- Calculate mean squares: $MSB = SSB/2$ and $MSW = SSW/9$. Compute F-statistic $F = MSB/MSW$ and compare to critical value 4.26. Is there significant variation between dosage groups at $\alpha = 0.05$? What does this mean for drug effectiveness?
- Estimate variance components. Within-group variance: $\hat{\sigma}_W^2 = MSW$. Between-group variance: $\hat{\sigma}_B^2 = (MSB - MSW)/4$. Calculate intraclass correlation $\rho_I = \hat{\sigma}_B^2 / (\hat{\sigma}_B^2 + \hat{\sigma}_W^2)$ which measures proportion of variance due to dosage. Interpret the result.
- Power analysis: To detect effect size $f = \sigma_B / \sigma_W = 0.5$ (medium) with power 0.80 at $\alpha = 0.05$, calculate required sample size per group using $n \approx 2(z_{\alpha/2} + z_\beta)^2 / [(a-1)f^2]$ where $a = 3$ groups. Is the current design ($n = 4$ per group) adequately powered?
- Blocking design: Using the two-way ANOVA sums of squares given, calculate: (a) Efficiency of blocking $= (SS_{\text{Age}} + SS_{\text{Error}}) / SS_{\text{Error}}$, showing how much larger error would be without blocking. (b) Proportion of variance explained by age $= SS_{\text{Age}} / (SS_{\text{Dosage}} + SS_{\text{Age}} + SS_{\text{Error}})$. Was age blocking effective?
- Sample size determination: For estimating mean response time at Medium dosage with margin of error $E = 2$ minutes at 95% confidence, using pilot $s = 4.5$, calculate $n = (z_{\alpha/2} \cdot s / E)^2$. For variance precision (CI width $\leq 50\%$ of 2σ), approximately $n \geq 40$ is needed. Which constraint (mean or variance) dominates the required sample size?